

Project: AgriLink

Farm Management & Agricultural Services Platform

Domain: Agriculture & Rural Development

1. Introduction

AgriLink is a web-based farm management and agricultural services platform designed for agricultural departments, farmer producer organisations, cooperative societies, and agri-input companies to manage farmer registrations, crop planning, input procurement, subsidy disbursements, produce sales, and rural advisory services. The system enables farmers to register land holdings, plan crop calendars, apply for subsidies, and connect with buyers; agri-extension officers to provide advisory services and monitor field progress; procurement officers to manage produce buy-back and price discovery; and administrators to configure schemes, crop calendars, and compliance rules.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Farmer:** Registers land, plans crop seasons, applies for subsidies, and lists produce for sale
- **Agri-Extension Officer:** Provides crop advisory, conducts field visits, and records observations
- **Procurement Officer:** Manages produce buy-back, price negotiation, and payment disbursements
- **Subsidy Administrator:** Reviews eligibility, approves schemes, and tracks disbursement status
- **Compliance Analyst:** Audits scheme utilisation, validates land records, and generates reports
- **AgriLink Admin:** Configures crop calendars, scheme catalog, input price lists, and user roles

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.

- **2.2 Farmer & Land Registration:** Manages farmer profiles, land holdings, and crop history.
- **2.3 Crop Planning & Season Management:** Manages crop calendars, sowing plans, and growth stage tracking.
- **2.4 Agri-Input & Procurement Management:** Manages seed, fertiliser, and pesticide procurement requests.
- **2.5 Subsidy & Scheme Management:** Handles subsidy applications, eligibility checks, and disbursements.
- **2.6 Produce Sales & Market Linkage:** Manages produce listings, buyer matching, and payment settlements.
- **2.7 Agricultural Analytics & Reporting:** Provides dashboards for crop coverage, yield estimates, and scheme utilisation.
- **2.8 Notifications & Alerts:** Crop advisory alerts, subsidy approval updates, and market price notifications.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Farmer and staff registration, login, and role-based access control
- Region-scoped session management and full audit logging

Entities

- **User**
 - UserID
 - Name
 - Role
(Farmer/ExtensionOfficer/ProcurementOfficer/SubsidyAdmin/Compliance/Admin)
 - Email

- Phone
- RegionID
- Status (Active/Inactive)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - Module
 - Timestamp

4.2 Farmer & Land Registration Module

Features

- Register farmers with identity details and land holding information
- Maintain crop history, soil type records, and irrigation source details

Entities

- **FarmerProfile**
 - FarmerID
 - Name
 - DateOfBirth
 - Gender
 - NationalIDNumber
 - Village
 - District
 - State
 - Phone
 - BankAccountNumber
 - Status (Active/Inactive/Verified)
- **LandHolding**

- HoldingID
- FarmerID
- SurveyNumber
- AreaAcres
- SoilType (Clay/Sandy/Loam/Black)
- IrrigationSource (Rain/Canal/Borewell/None)
- OwnershipType (Owned/Leased/SharedCropping)
- Status (Active/Disputed)

4.3 Crop Planning & Season Management Module

Features

- Define seasonal crop calendars and assign crops to land holdings
- Track sowing, growth stage milestones, and expected harvest timelines

Entities

- **CropCatalog**
 - CropID
 - CropName
 - Category (Cereal/Pulse/Vegetable/Fruit/Cash)
 - Season (Kharif/Rabi/Zaid/Perennial)
 - TypicalDurationDays
 - ExpectedYieldPerAcre
 - Status (Active/Inactive)
- **CropPlan**
 - PlanID
 - FarmerID
 - HoldingID
 - CropID
 - Season

- Year
- SowingDate
- ExpectedHarvestDate
- AreaPlanted
- Status (Planned/Sowing/Growing/Harvested/Failed)
- **GrowthObservation**
 - ObservationID
 - PlanID
 - OfficerID
 - ObservationDate
 - Stage (Germination/Vegetative/Flowering/Maturity)
 - PestOrDiseaseFlag
 - Remarks

4.4 Agri-Input & Procurement Management Module

Features

- Manage seed, fertiliser, and pesticide catalog with pricing
- Process farmer input procurement requests and delivery tracking

Entities

- **InputCatalog**
 - InputID
 - Name
 - Category (Seed/Fertiliser/Pesticide/Equipment)
 - Unit
 - PricePerUnit
 - SubsidisedPrice
 - AvailableStock
 - Status (Available/OutOfStock)

- **InputRequest**
 - RequestID
 - FarmerID
 - InputID
 - QuantityRequested
 - RequestDate
 - AssignedCentreID
 - ActualPrice
 - Status (Requested/Approved/Dispatched/Delivered/Cancelled)

4.5 Subsidy & Scheme Management Module

Features

- Configure government scheme catalog with eligibility criteria and benefit amounts
- Process farmer applications, verify eligibility, and track disbursements

Entities

- **SchemeCatalog**
 - SchemeID
 - SchemeName
 - Category (InputSubsidy/CropInsurance/EquipmentGrant/WelfareSupport)
 - EligibilityCriteria
 - BenefitAmount
 - FundingSource
 - StartDate
 - EndDate
 - Status (Active/Closed/Upcoming)
- **SubsidyApplication**
 - ApplicationID
 - FarmerID

- SchemeID
- ApplicationDate
- EligibilityScore
- ReviewedBy
- DisbursedAmount
- DisbursedDate
- Status (Submitted/UnderReview/Approved/Rejected/Disbursed)

4.6 Produce Sales & Market Linkage Module

Features

- List harvested produce with quantity, quality grade, and asking price
- Match produce listings with registered buyers and track payment settlements

Entities

- **ProduceListing**
 - ListingID
 - FarmerID
 - CropID
 - HarvestDate
 - QuantityKg
 - QualityGrade (A/B/C)
 - AskingPricePerKg
 - Status (Available/PartiallyBooked/Sold/Withdrawn)
- **ProduceSale**
 - SaleID
 - ListingID
 - BuyerID
 - QuantitySoldKg
 - AgreedPricePerKg

- TotalAmount
- SaleDate
- PaymentStatus (Pending/Paid/Overdue)

4.7 Agricultural Analytics & Reporting Module

Features

- Crop coverage area, yield estimate, and harvest volume dashboards by crop and region
- Subsidy scheme utilisation, input procurement trends, and produce sales performance reports

Entities

- **AgriReport**
 - ReportID
 - Scope (District/Crop/Season/Scheme)
 - Metrics (RegisteredFarmers, TotalAreaCovered, EstimatedYield, SubsidyDisbursed, ProduceSalesValue, InputDeliveryRate)
 - GeneratedDate

4.8 Notifications & Alerts Module

Features

- Sowing reminders, pest alert advisories, and harvest deadline notifications for farmers
- Subsidy disbursement confirmations, input delivery updates, and scheme closure alerts

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (CropAdvisory/Subsidy/InputProcurement/ProduceSale/Compliance)

- Status (Unread/Read/Dismissed)
- CreatedDate

5. Deployment Strategy

- Local: Angular/React frontend, Spring Boot/.NET Core backend, local DB
- Production: Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, RegionID, Status)
- **AuditLog**(AuditID, UserID, Action, Module, Timestamp)
- **FarmerProfile**(FarmerID, Name, DateOfBirth, Gender, NationalIDNumber, Village, District, State, Phone, BankAccountNumber, Status)
- **LandHolding**(HoldingID, FarmerID, SurveyNumber, AreaAcres, SoilType, IrrigationSource, OwnershipType, Status)
- **CropCatalog**(CropID, CropName, Category, Season, TypicalDurationDays, ExpectedYieldPerAcre, Status)
- **CropPlan**(PlanID, FarmerID, HoldingID, CropID, Season, Year, SowingDate, ExpectedHarvestDate, AreaPlanted, Status)
- **GrowthObservation**(ObservationID, PlanID, OfficerID, ObservationDate, Stage, PestOrDiseaseFlag, Remarks)
- **InputCatalog**(InputID, Name, Category, Unit, PricePerUnit, SubsidisedPrice, AvailableStock, Status)
- **InputRequest**(RequestID, FarmerID, InputID, QuantityRequested, RequestDate, AssignedCentreID, ActualPrice, Status)
- **SchemeCatalog**(SchemeID, SchemeName, Category, EligibilityCriteria, BenefitAmount, FundingSource, StartDate, EndDate, Status)
- **SubsidyApplication**(ApplicationID, FarmerID, SchemeID, ApplicationDate, EligibilityScore, ReviewedBy, DisbursedAmount, DisbursedDate, Status)
- **ProduceListing**(ListingID, FarmerID, CropID, HarvestDate, QuantityKg, QualityGrade, AskingPricePerKg, Status)
- **ProduceSale**(SaleID, ListingID, BuyerID, QuantitySoldKg, AgreedPricePerKg, TotalAmount, SaleDate, PaymentStatus)

- **AgriReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Farmer Portal:** Land and crop dashboard, input request tracker, subsidy application status, produce listing manager
- **Extension Officer Console:** Assigned farmers list, field visit log, growth observation form, crop advisory bulletin board
- **Procurement Dashboard:** Active produce listings, buyer matching queue, sale settlement tracker, price discovery panel
- **Subsidy Administration Console:** Scheme catalog, application review queue, eligibility checker, disbursement log
- **Analytics Dashboard:** Crop coverage map by region, yield trend charts, subsidy utilisation tracker, produce sales summary
- **Admin Console:** Crop and scheme catalog builder, input price configuration, region and officer assignment

8. Non-Functional Requirements

- **Performance:** Support 20,000+ concurrent farmer and officer users during sowing season registration peaks
- **Security:** Farmer PII protection, bank account data encryption, RBAC, and full audit trails for disbursements
- **Scalability:** Multi-district and multi-state deployment with region-scoped data partitioning
- **Availability:** 99.9% uptime with offline-ready form capture for field officers in low-connectivity areas
- **Maintainability:** Configurable scheme eligibility rules, crop calendars, and input price lists without code changes
- **Observability:** Seasonal crop dashboards, subsidy pipeline metrics, and produce market price trend monitoring

9. Assumptions & Constraints

- Phase 1 excludes integration with national land registry APIs and external weather data feeds

- Market price discovery uses internally maintained price lists; commodity exchange integration is deferred
- All notifications are in-app only; SMS advisory integrations for low-smartphone farmers are out of Phase 1 scope
- Soil testing results and remote sensing data integration are deferred to Phase 2
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling